

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	
Project Title	Rising Waters - ML-Based Flood Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform flood-risk assessment using machine learning, enabling faster and more reliable predictions than manual analysis. It tackles the lack of an accessible, data-driven early-warning tool, promising quicker response, reduced damage, and safer communities. Key features include a trained classification model and an easy-to-use web interface for real-time flood prediction.

Project Overview	
Objective	The primary objective is to accurately predict the likelihood of flooding in a given region by applying machine learning techniques to rainfall and weather data, enabling faster and more informed decisions.
Scope	The project covers data collection and preprocessing, exploratory analysis, training and comparing multiple classification models, and deploying the best-performing model through a Flask web application for real-time flood prediction.

Problem Statement	
Description	Flood-prone regions often lack a quick, localized way to assess flood risk from current rainfall and weather conditions, leading to delayed warnings and reactive rather than preventive responses.
Impact	Solving this will enable earlier, data-backed flood warnings, giving authorities and residents more time to prepare, reducing loss of life, property damage, and disruption caused by floods.

Proposed Solution	
Approach	Employing machine learning classification techniques trained on historical rainfall and weather data to predict flood occurrence, packaged as a web application for real-time, accessible predictions.
Key Features	- Implementation of a machine learning-based flood prediction model (Decision Tree, Random Forest, KNN, and XGBoost compared).
	- Selection of the best-performing model (XGBoost, 95.65% test accuracy) for deployment.- Real-time flood-risk prediction through a simple Flask web interface.- Uses historical India rainfall data (1901-2015) alongside a labelled flood dataset for training and validation.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (no GPU required)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	5 GB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	numpy, pandas, matplotlib, seaborn, scikit-learn, xgboost, joblib, openpyxl, gunicorn
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Historical India rainfall dataset (1901-2015), xlsx Labelled flood dataset, 115 records, xlsx